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B032426 (B255) - ARCHITECTURE, MODEL AND ANALYSIS OF CYBER PHYSICAL SYSTEMS

# Bayesian Networks

## Lecture 3

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# Recap from last lecture

You learned how to:

- Justify the usage of Bayesian Networks in terms of ease of use, solving efficiency, and expressive power compared to Fault Trees and Petri Nets
- Translate a Fault Tree model into a Bayesian Network model
- Enrich the BN model with further relevant details such as common mode of failures
- Implement and evaluate BN models of fault-tolerant systems starting from FT models

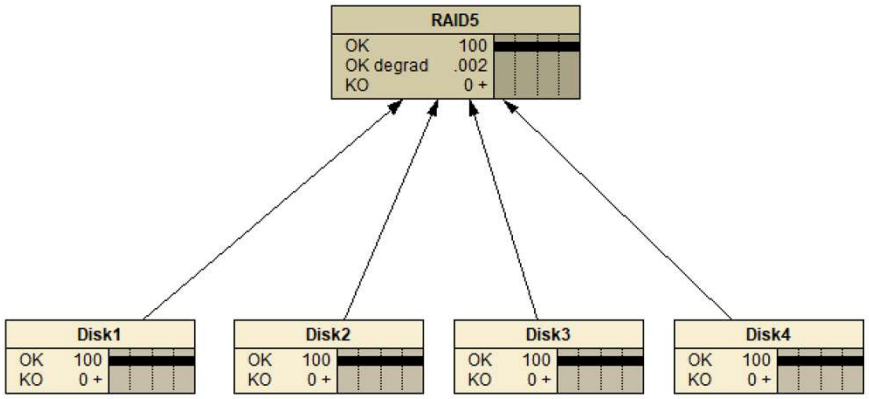
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# Home assignment: RAID 5 storage system

- Model the following failure modes of a RAID 5 storage system with 4 units: 1) working at full performance; 2) working at degraded performance due to 1 unit failure; 3) failure.
- Consider a single unit MTBF of 200.000h. Start from a FT, translate it to a BN, and then add and model the third failure mode.
- Implement and evaluate your model in Netica.
- Prepare a few slides Powerpoint presentation about the model and results to be shown and discussed during next lecture.
- Assignment is for learning purposes only, and therefore not graded.
- Working in small groups is allowed.

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# RAID5 BN model exercise




Node: RAID5 Apply

Deterministic Function Reset

| Disk1 | Disk2 | Disk3 | Disk4 | RAID5 |
|---|---|---|---|---|
| OK | OK | OK | OK | OK |
| OK | OK | OK | KO | OK degrad |
| OK | OK | KO | OK | OK degrad |
| OK | OK | KO | KO | KO |
| OK | KO | OK | OK | OK degrad |
| OK | KO | OK | KO | KO |
| OK | KO | KO | OK | KO |
| OK | KO | KO | KO | KO |
| KO | OK | OK | OK | OK degrad |
| KO | OK | OK | KO | KO |
| KO | OK | KO | OK | KO |
| KO | OK | KO | KO | KO |
| KO | KO | OK | OK | KO |
| KO | KO | OK | KO | KO |
| KO | KO | KO | OK | KO |
| KO | KO | KO | KO | KO |

```
Disk1
OK    0.99999
KO    5e-06

Disk2
OK    0.99999
KO    5e-06

Disk3
OK    0.99999
KO    5e-06

Disk4
OK    0.99999
KO    5e-06

RAID5
OK            0.99998
OK_degrad     2e-05
KO            1.5e-10
```

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# Learning objectives of this lecture

At the end of this lecture you will be able to:

- Describe the architecture, functionalities, and main failure modes of a modern computer-based railway control system
- Combine multiple modeling formalism to model different aspects/subsystems
- Leverage on the BN formalism for modeling main failure-modes

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# Learning materials for this lecture

F. Flammini, S. Marrone, N. Mazzocca, and V. Vittorini, "Modelling system reliability aspects of ERTMS/ETCS by Fault Trees and Bayesian Networks," in *Proc. European Safety and Reliability Conference, ESREL*, 2006, pp. 2675–2683.

Free full-text download from:

https://www.researchgate.net/publication/232560252_Modelling_system_reliability_aspects_of_ERTMSETCS_by_Fault_Trees_and_Bayesian_Networks

*Safety and Reliability for Managing Risk – Guedes Soares & Zio (eds)*
*© 2006 Taylor & Francis Group, London, ISBN 0-415-41620-5*

Modeling system reliability aspects of ERTMS/ETCS by fault trees and Bayesian networks

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[2]*Università di Napoli "Federico II" – Dipartimento di Informatica e Sistemistica, Naples, Italy*
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ABSTRACT: Critical control systems require proper techniques to predict their failure rate since early design stages, in order to fulfil dependability requirements and minimize development costs. Bayesian Networks have been shown to be suitable to model system reliability aspects, extending the modeling power of Fault Trees and featuring a better solving efficiency with respect to Petri Nets. In this paper we exploit the Fault Tree and Bayesian Network formalisms in order to perform a hardware reliability analysis of a complex real world case study: the European Railway Traffic Management System/European Train Control System (ERTMS/ETCS). ERTMS/ETCS is a recent standard specification aimed at improving interoperability, performances and dependability of modern railways. An implementation of ERTMS/ETCS is a distributed heterogeneous system with strict reliability requirements. Starting from such requirements and from a reference hardware architecture, we studied system reliability by instantiating models with realistic parameters and performing a series of sensitivity analyses in order to highlight design trade-offs. By evaluating and integrating sub-models using a compositional approach we both obtained several interesting results and showed the effectiveness of a combined use of Fault Trees and Bayesian Networks in dealing with system reliability analyses of train control systems.

## 1 INTRODUCTION AND RELATED WORKS

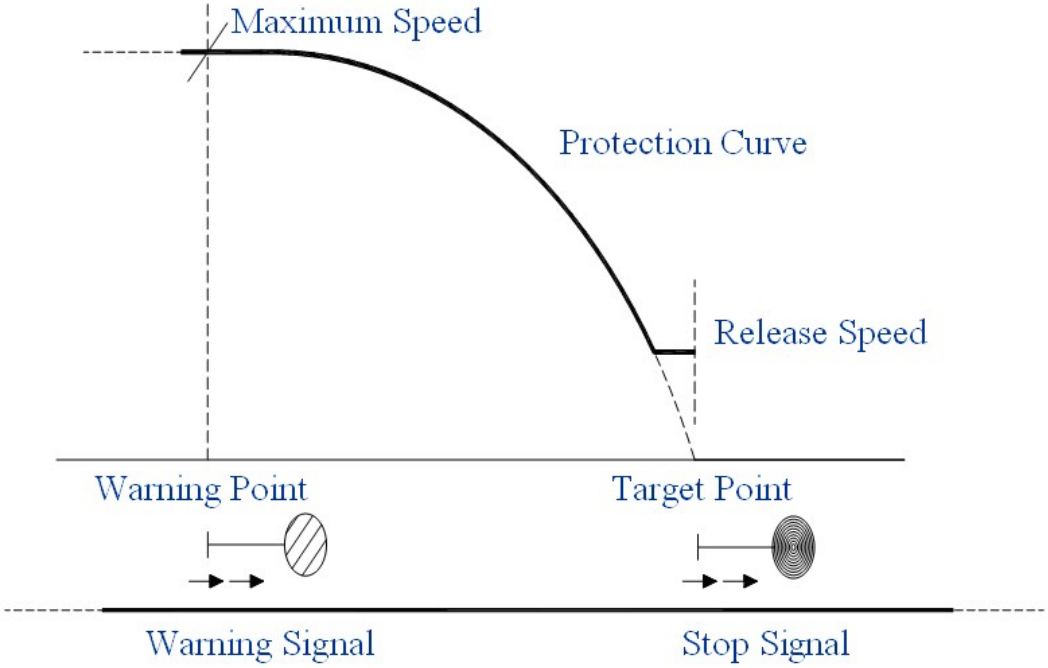
International RAMS (Reliability Availability Maintainability Safety) standards, e.g. CENELEC (CENELEC 1999), specifically address the techniques used to evaluate system dependability for critical control systems, giving general guidelines. The use of formal modeling techniques is highly recommended by such standards. Formal methods allow to predict system reliability since early stages of system development, reducing the probability of design reviews. The modeling language should be chosen with the aim of balancing easy of use, expressive power and solving efficiency. Several formal languages and methods have been proposed by the scientific community in order to model system reliability aspects. Among them, Fault Trees (FT) and Reliability Block Diagrams (RBD) are limited in expressive power, but they are very efficient and easy to use; Continuous Time Markov Chains (CTMC) and the various kinds of Stochastic Petri Nets (SPN) allow to model any complex structure or behavior, but are usually not compatible with the complexity of very large systems, as their solving algorithms suffer from the state space explosion problem. This is also true for Fault Tree extensions which are solved by translating the extended FT model into a CTMC or SPN model (e.g. Dynamic and/or Repairable Fault Trees, see (Dugan 1992) and (Flammini 2005)). Finally, Bayesian Networks (BN) have been recently shown to be able to balance expressive power and solving efficiency in order to model system reliability aspects. BN and their extensions (e.g. Dynamic Bayesian Networks, see (Montani 2005)) provide a unified framework which is able to model nearly all reliability related issues. BNs have also been show to augment the expressive power of Fault Trees and achieve better efficiency with respect to Petri Nets and their extensions (e.g. GSPN, see (Ajmone 1995)). The BN formalism supports multi-state events, noisy gates, common mode failures, decision extensions and can be used to detect reliability bottlenecks and to diagnose failure causes starting from observable symptoms (the "evidence"). A methodology to translate a FT into a BN and a performance comparison among FT, PN and GSPN are presented in (Bobbio 2001b) and (Bobbio 2001a) respectively.

ERTMS/ETCS (European Railway Traffic Management System/European Train Control System) is a European standard specification aimed at improving safety, reliability, performance and interoperability of European railway lines (UNISIG 2002). In this paper, we exploit the Fault Tree and Bayesian Networks

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# Automatic Train Control (ATC) Systems

- Automatic Train Control (ATC) = Automatic Train Protection (ATP) + Automatic Train Operation (ATO)
- On-board elaboration of a Braking Curve or Dynamic Speed Profile
- Maximum Speed is the given by the Most Restrictive Speed Profile (MRSP) among all the Static Speed Profiles (SSP) and Temporary Speed Restrictions (TSR)
- Target Point (i.e. End of Authority, EoA) is obtained from the Movement Authority (MA)



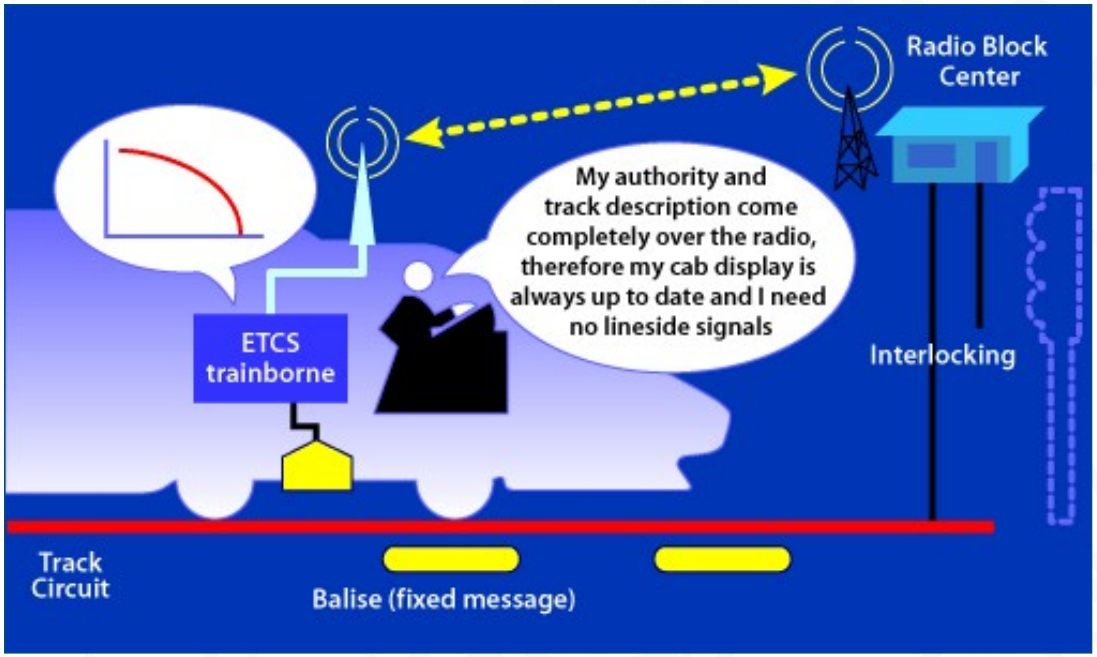
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# ERMTS/ETCS Level 2

- Continuous Communication Bases Signalling, based on:

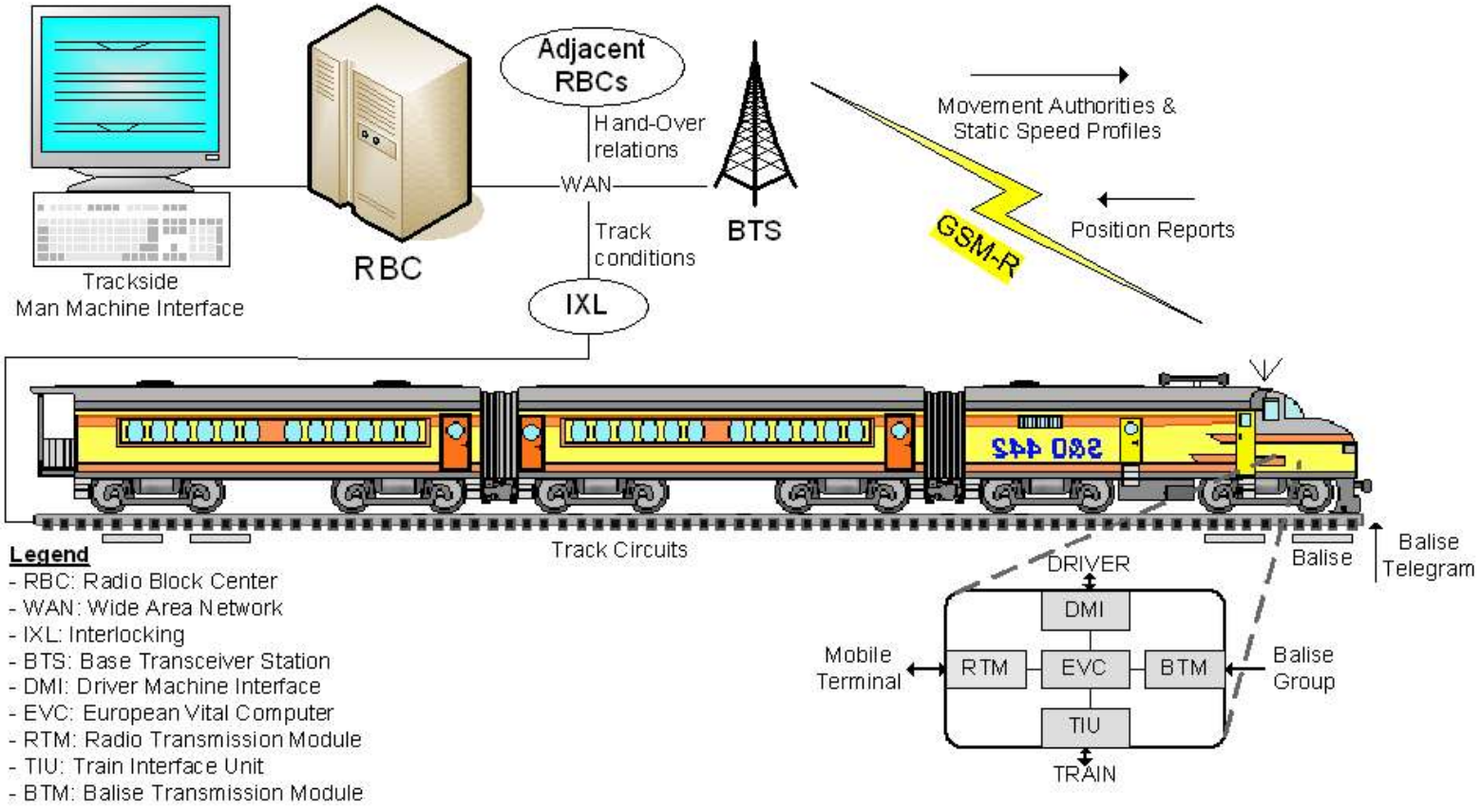
  *Global System for Mobile communications – Railways (GSM-R)*

- Light signals no more necessary
- Fixed message Balises used as *milestones*



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# ERTMS/ETCS



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## ERTMS failure modes

- Immobilizing Failure (IF): a failure which causes two or more trains to be switched into on-sight mode (i.e. they are no more under full system supervision);
- Service Failures (SF): a failure which causes at most one train to be switched into on-sight mode;
- Minor Failure (MF): a failure that results in unscheduled maintenance and cannot be classified in the above defined failure conditions.

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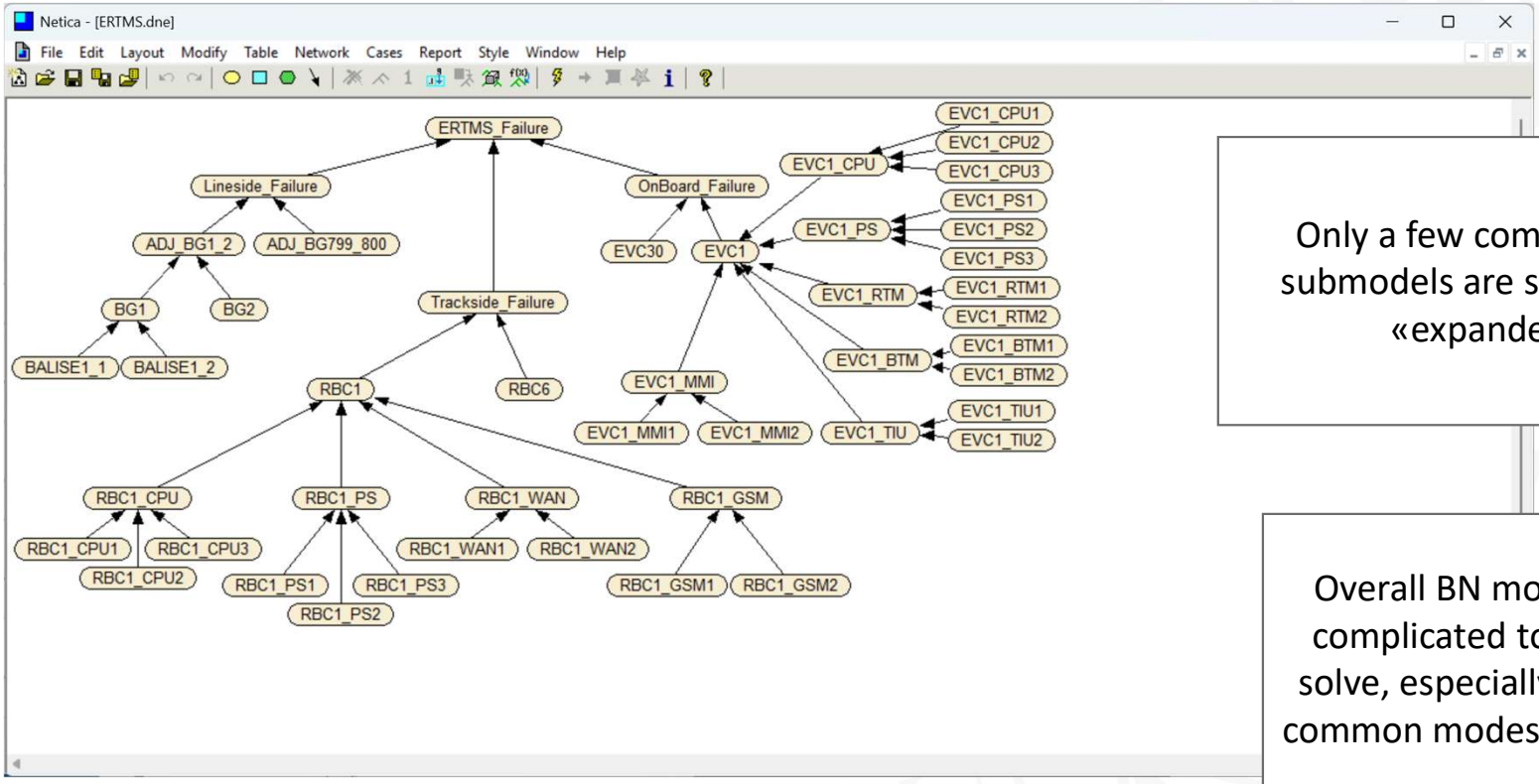
# ERTMS RAMS requirements

Table 1. ERTMS/ETCS RAM requirements of interest.

| Param. | Description | Value |
|---|---|---|
| MTBF-$I_{ONB}$ | MTBF w.r.t. IFs due to the On-board | $>2.7*10^6$ h |
| MTBF-$S_{ONB}$ | MTBF w.r.t. SFs due to the On-board | $>3*10^5$ h |
| MTBF-$I_{TRK}$ | MTBF w.r.t. IFs due to the Trackside | $>3.5*10^8$ h |
| MTBF-$I_{LNS}$ | MTBF w.r.t. IFs due to the Lineside | $>1.2*10^5$ h |
| $U_{RBC}$ | RBC Unavailability (*) | $<10^{-6}$ |
| $U_{BAL}$ | Balise Unavailability (*) | $<10^{-7}$ |
| $A_{IF-HW}$ | System Availability w.r.t. hardware IF | >0.999985 |
| $A_{SF-HW}$ | System Availability w.r.t. hardware SF | >0.99987 |

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# Example overall «template» BN for ERTMS failure in Netica

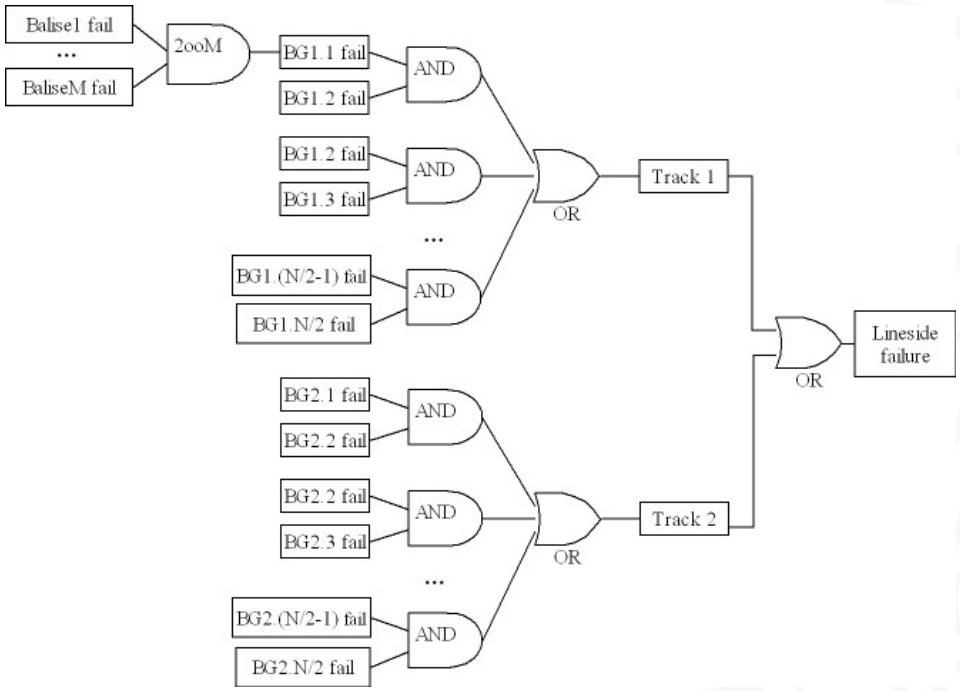



Only a few components' submodels are shown and «expanded»

Overall BN model too much complicated to manage and solve, especially when adding common modes of failures, etc.

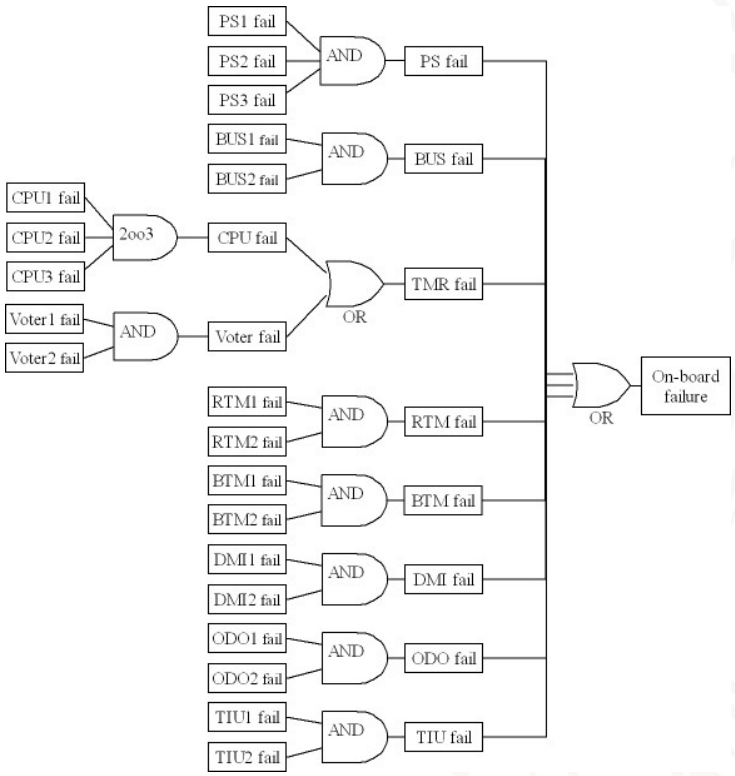
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# FT for the lineside failure



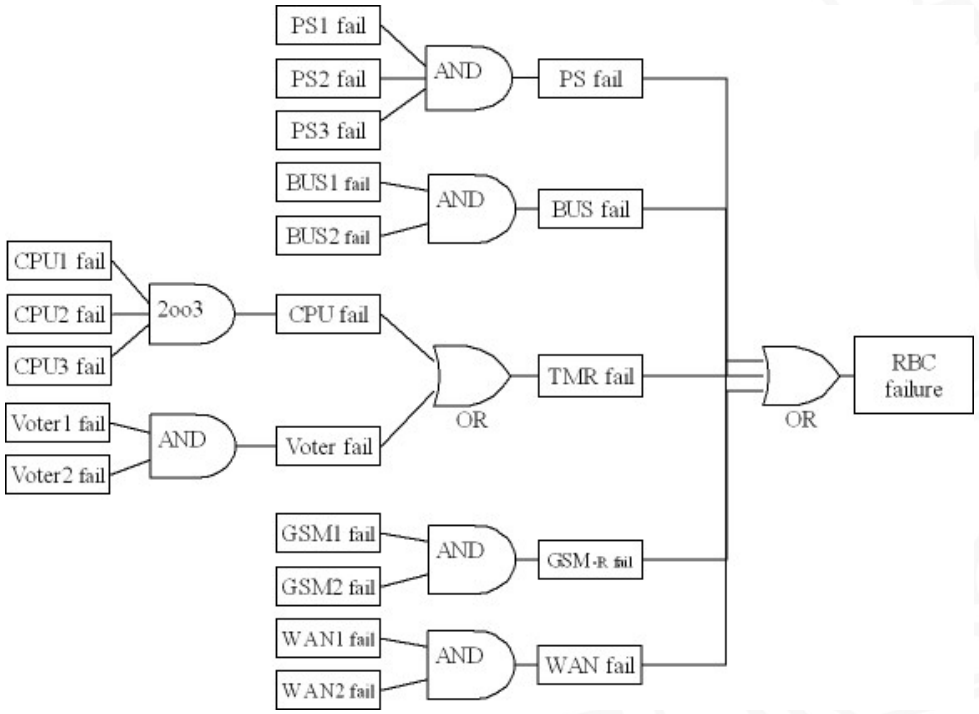
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# FT for the on-board failure



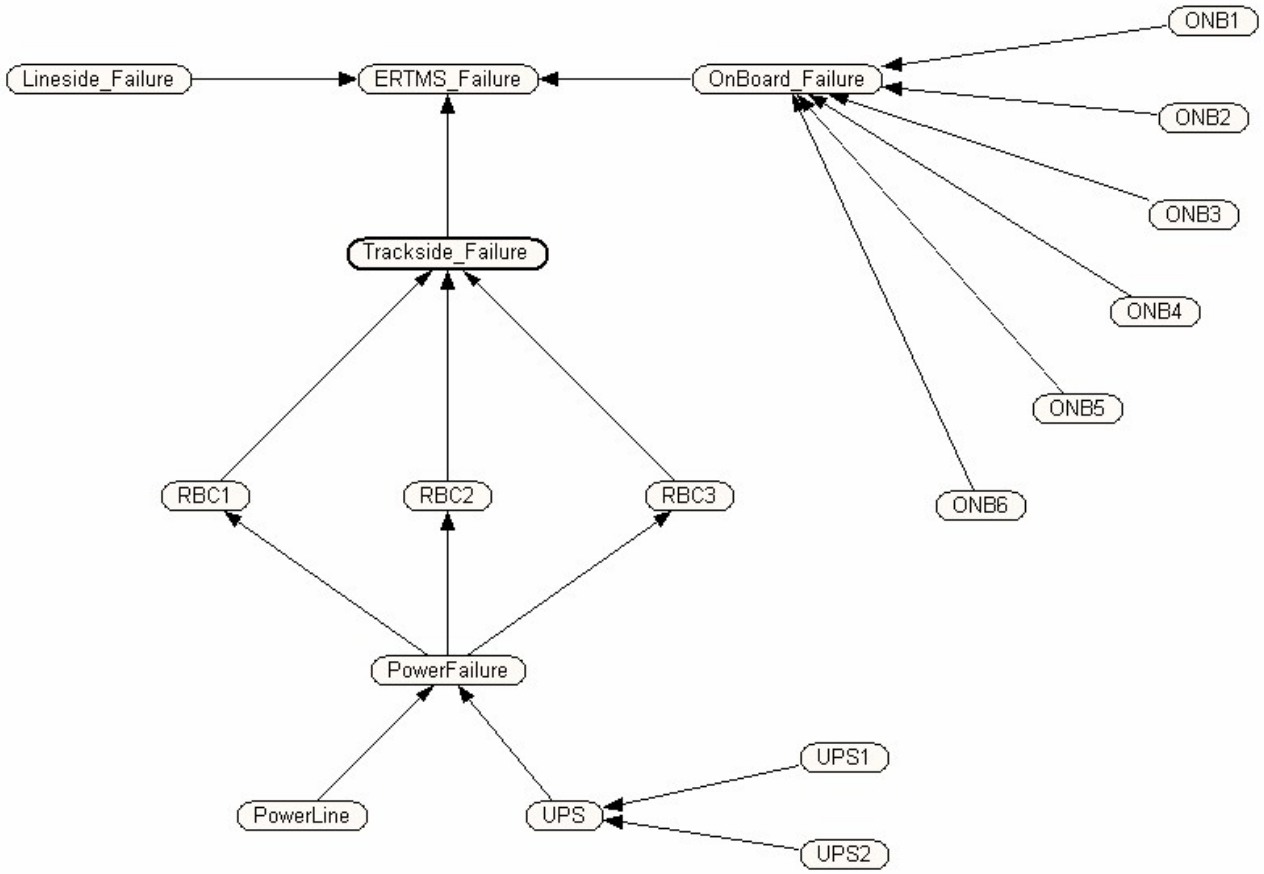
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# FT for the RBC failure



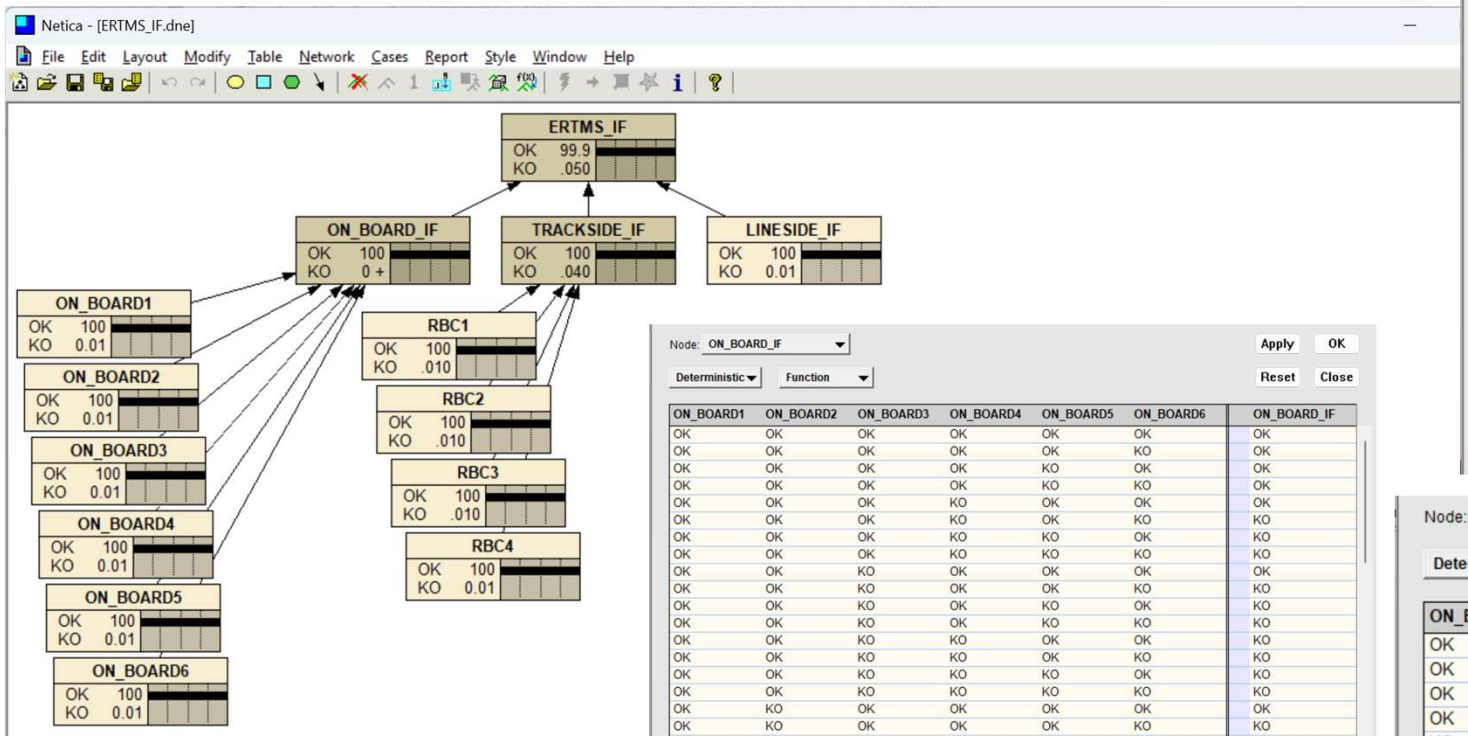
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# BN for the ERTMS failure with a common mode of failure



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# Example ERTMS Immobilizing Failure BN model in Netica




Node: ON_BOARD_IF — Apply — OK — Deterministic — Function — Reset — Close

| ON_BOARD1 | ON_BOARD2 | ON_BOARD3 | ON_BOARD4 | ON_BOARD5 | ON_BOARD6 | ON_BOARD_IF |
|---|---|---|---|---|---|---|
| OK | OK | OK | OK | OK | OK | OK |
| OK | OK | OK | OK | OK | KO | OK |
| OK | OK | OK | OK | KO | OK | OK |
| OK | OK | OK | OK | KO | KO | OK |
| OK | OK | OK | KO | OK | OK | OK |
| OK | OK | OK | KO | OK | KO | KO |
| OK | OK | OK | KO | KO | OK | KO |
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| OK | KO | KO | KO | OK | OK | KO |
| OK | KO | KO | KO | OK | KO | KO |
| OK | KO | KO | KO | KO | OK | KO |
| OK | KO | KO | KO | KO | KO | KO |

Node: TRACKSIDE_IF — Apply — OK — Deterministic — Function — Reset — Close

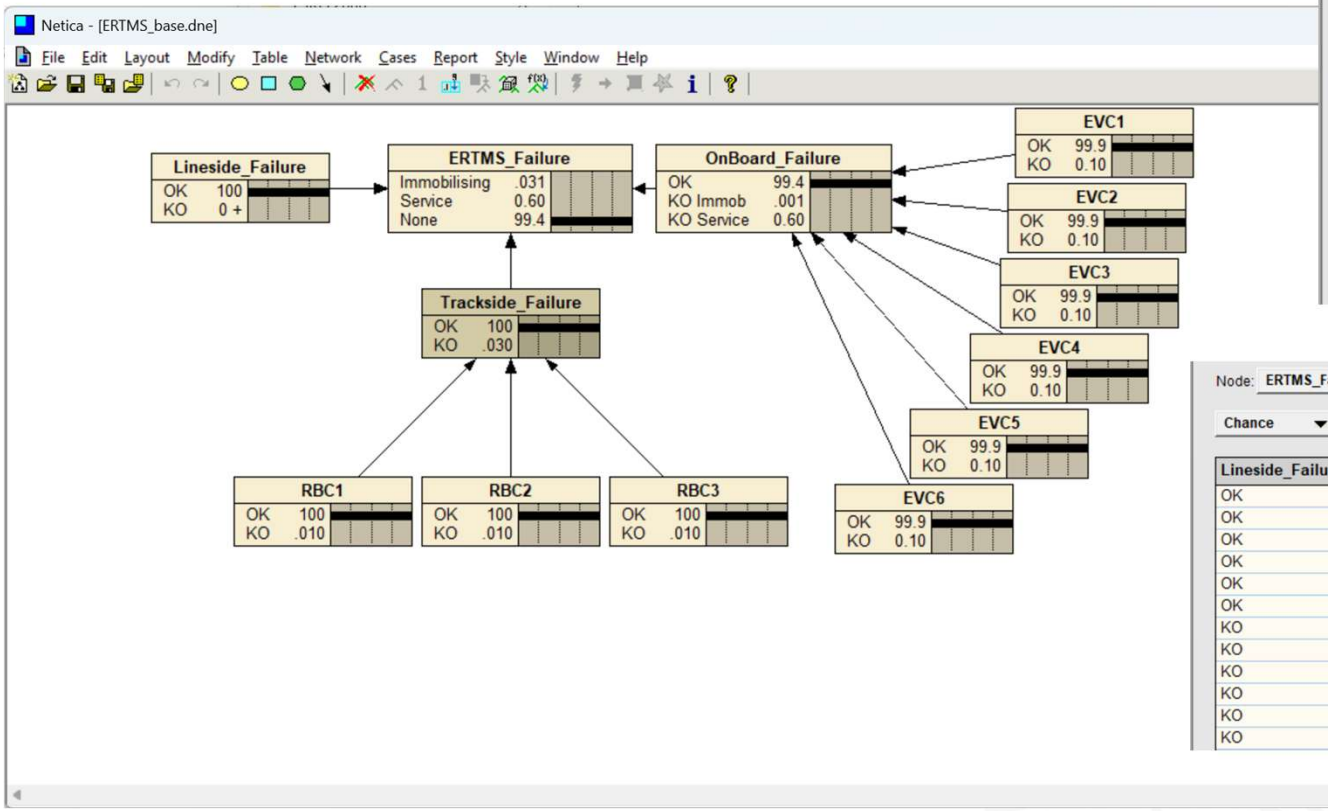
| RBC1 | RBC2 | RBC3 | RBC4 | TRACKSIDE_IF |
|---|---|---|---|---|
| OK | OK | OK | OK | OK |
| OK | OK | OK | KO | KO |
| OK | OK | KO | OK | KO |
| OK | OK | KO | KO | KO |
| OK | KO | OK | OK | KO |
| OK | KO | OK | KO | KO |
| OK | KO | KO | OK | KO |
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| KO | OK | OK | OK | KO |
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| KO | OK | KO | KO | KO |
| KO | KO | OK | OK | KO |
| KO | KO | OK | KO | KO |
| KO | KO | KO | OK | KO |
| KO | KO | KO | KO | KO |

Node: IF — Apply — OK — Deterministic — Function — Reset — Close

| ON_BOARD_IF | LINESIDE_IF | TRACKSIDE_IF | ERTMS_IF |
|---|---|---|---|
| OK | OK | OK | OK |
| OK | OK | KO | KO |
| OK | KO | OK | KO |
| OK | KO | KO | KO |
| KO | OK | OK | KO |
| KO | OK | KO | KO |
| KO | KO | OK | KO |
| KO | KO | KO | KO |

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# ERTMS BN failure model in Netica




Node: Trackside_Failure — Apply

Deterministic — Function — Reset

| RBC1 | RBC2 | RBC3 | Trackside_Failure |
|---|---|---|---|
| OK | OK | OK | OK |
| OK | OK | KO | KO |
| OK | KO | OK | KO |
| OK | KO | KO | KO |
| KO | OK | OK | KO |
| KO | OK | KO | KO |
| KO | KO | OK | KO |
| KO | KO | KO | KO |

Node: ERTMS_Failure — Apply — OK

Chance — % Probability — Reset — Close

| Lineside_Failure | Trackside_Failure | OnBoard_Failure | Immobilising | Service | None |
|---|---|---|---|---|---|
| OK | OK | OK | | | 100 |
| OK | OK | KO Immob | 100 | | |
| OK | OK | KO Service | | 100 | |
| OK | KO | OK | 100 | | |
| OK | KO | KO Immob | 100 | | |
| OK | KO | KO Service | 100 | | |
| KO | OK | OK | 100 | | |
| KO | OK | KO Immob | 100 | | |
| KO | OK | KO Service | 100 | | |
| KO | KO | OK | 100 | | |
| KO | KO | KO Immob | 100 | | |
| KO | KO | KO Service | 100 | | |

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# Assignment: ERTMS simplified

- Using the parameters from the paper when necessary/appropriate, build a BN model of a simplified ERTMS track with only 1 RBC and 3 trains (read the paper for further details and parameters)
- Implement and evaluate your model in Netica.
- Prepare a few slides Powerpoint presentation about the model and results to be shown and discussed during next lecture.
- Working in small groups is allowed.
- Start working in classroom, then continue at home.
- Assignment is for learning purposes only, and therefore not graded.

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# Next lecture

- Quick recap and questions about today's lecture

- Presentation of BN models developed by students

- BN safety modelling of N-modular redundant computer architectures

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# Thank you for your attention

## Questions?

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